

KOKSHETAU, Kazakhstan

WMO: 288790

Lat: 53.3169N

Lon: 69.3772E

Elev: 751

StdP: 14.30

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-19.9	-32.1	0.9	-26.3	-25.6	1.4	-19.6	34.6	17.5	32.0	20.8	3.6	250	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	87.9	65.1	84.7	64.7	81.3	63.7	68.8	81.2	67.2	79.4	65.8	77.3	7.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	94.1	73.2	63.0	88.7	71.9	61.4	83.5	71.0	33.4	81.3	32.2	79.2	31.0	77.2	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.6	26.1	21.0	-30.8	92.5	7.0	3.5	-35.8	95.0	-39.9	97.1	-43.9	99.1	-48.9	101.6		
(5)			-31.2	71.9	7.6	2.3	-36.7	73.6	-41.1	75.0	-45.3	76.3	-50.8	78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	4.9	7.4	22.3	41.6	55.5	65.3	66.9	64.4	54.0	40.6	22.9	10.6
	DBStd	24.84	12.91	12.78	10.97	10.45	9.00	7.70	6.54	7.63	8.73	8.96	13.65	12.75
	HDD50	6193	1399	1192	860	287	50	3	0	2	54	313	813	1220
	HDD65	10115	1864	1612	1325	702	312	88	56	106	346	756	1263	1685
	CDD50	1888	0	0	0	36	220	463	525	449	173	22	0	0
	CDD65	333	0	0	0	0	17	98	116	88	14	0	0	0
	CDH74	3262	0	0	0	9	236	971	1071	820	155	0	0	0
	CDH80	976	0	0	0	1	51	310	321	250	43	0	0	0
(14) Wind	WSAvg	8.9	8.1	9.3	10.1	9.6	9.8	8.2	7.3	7.3	8.2	9.4	10.1	9.9
(15) Precipitation	PrecAvg	12.2	0.4	0.4	0.4	0.7	1.1	1.5	2.9	1.8	0.9	0.8	0.6	0.4
	PrecMax	20.6	1.9	1.6	1.1	2.4	2.4	4.1	7.7	4.8	2.6	2.9	2.4	1.3
	PrecMin	7.9	0.0	0.0	0.0	0.0	0.0	0.1	0.8	0.1	0.3	0.0	0.0	0.0
	PrecStd	3.5	0.4	0.3	0.3	0.5	0.7	0.9	1.9	1.1	0.6	0.6	0.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	34.9	49.7	74.0	85.4	91.0	92.2	90.1	85.5	69.4	48.2	35.4
		MCWB	32.8	32.4	41.2	54.7	61.0	64.6	67.0	66.4	62.2	52.9	42.6	32.7
	2%	DB	30.6	31.6	42.1	67.8	79.8	87.2	87.3	86.3	78.2	63.6	44.1	32.9
		MCWB	28.8	29.5	36.9	52.0	59.2	64.3	66.2	65.6	58.4	50.4	40.0	31.1
	5%	DB	27.1	29.0	38.7	62.4	75.8	83.7	83.2	82.6	72.8	58.4	40.8	30.5
		MCWB	25.5	26.9	34.7	49.0	57.6	64.2	65.6	64.7	57.0	47.7	37.6	28.5
	10%	DB	23.8	25.5	36.0	57.3	71.9	79.7	79.8	78.6	67.9	53.8	37.9	27.4
		MCWB	22.2	23.5	33.2	46.7	56.2	62.5	65.1	63.2	54.8	45.8	34.9	25.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.1	33.0	41.6	56.1	64.8	69.8	70.9	70.2	64.1	54.2	44.4	33.3
		MCDB	34.6	34.8	48.0	70.1	79.4	83.6	82.1	83.5	80.6	65.5	47.6	34.6
	2%	WB	28.8	29.8	37.4	53.0	61.6	67.3	69.2	67.9	60.5	51.7	40.3	31.2
		MCDB	30.5	31.6	41.6	64.7	75.3	81.1	80.7	80.8	73.8	61.1	43.5	32.9
	5%	WB	25.5	27.1	35.2	50.4	59.2	65.6	67.8	66.2	58.2	48.7	37.8	28.6
		MCDB	26.9	28.9	38.2	60.8	73.1	78.9	79.6	78.4	70.2	56.8	40.7	30.5
	10%	WB	22.2	23.6	33.3	47.5	57.3	64.0	66.3	64.5	55.9	46.2	35.0	25.5
		MCDB	23.8	25.4	36.1	56.9	69.4	76.3	76.8	76.1	66.3	53.0	37.6	27.4
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	15.1	15.1	17.2	20.4	19.5	18.5	19.1	17.4	13.5	11.1	12.5
		MCDBR	11.8	12.1	14.4	25.0	26.1	24.5	24.2	24.8	24.8	22.3	11.9	12.5
		MCWBR	11.6	10.9	10.6	13.0	12.3	10.1	9.7	10.5	11.5	12.6	9.3	11.6
	5% WB	MCDBR	11.9	11.6	13.2	23.2	23.5	21.2	20.3	22.0	21.4	20.3	11.2	12.2
		MCWBR	11.7	10.6	10.3	12.9	12.5	9.8	10.0	10.6	11.2	12.1	9.3	11.7
(40) Clear-Sky Solar Irradiance	taub		0.283	0.312	0.369	0.406	0.404	0.407	0.429	0.421	0.367	0.346	0.318	0.283
	taud		2.076	2.042	1.989	2.139	2.222	2.293	2.221	2.227	2.332	2.282	2.148	2.065
	Ebn at Noon		209	241	250	260	268	268	259	253	254	231	193	181
	Edh at Noon		23	33	43	43	42	39	42	39	31	26	22	19
(44) All-Sky Solar Radiation	RadAvg		370	725	1190	1468	1792	1914	1761	1506	1082	571	285	247
	RadStd		26	30	131	152	150	153	136	123	79	87	41	31

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page